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Terang

Satellite electrification intelligence

Hackathon Proposal

Executive Summary

While Indonesia reports an official village electrification ratio of 99.90%, national utility figures reveal that only 92.75% of villages are connected to the grid. The remaining gap is filled by unmonitored solar lanterns or small diesel systems that quickly fail, leaving over 4,808 unconnected villages—concentrated primarily across eastern Indonesia and archipelagic ASEAN—invisible to planners, utilities, and investors. While macro-initiatives like ASEAN Power Grid (APG) focus on interconnecting national grids across mainlands, Terang addresses the millions of people in island communities that exist beyond the physical and economic reach of any grid.

We have tested this approach in Maluku province, analyzing all 1,233 villages using free satellite nighttime-light imagery. By replacing standard built-up pixel masking with an unmasked polygon-based reduction—recovering the 79% of remote settlements typically lost in standard geospatial analyses—we identified 111 settled villages emitting light at or below uninhabited baseline levels, alongside 894 villages suffering from severe, seasonal service intermittency (Figure 1).

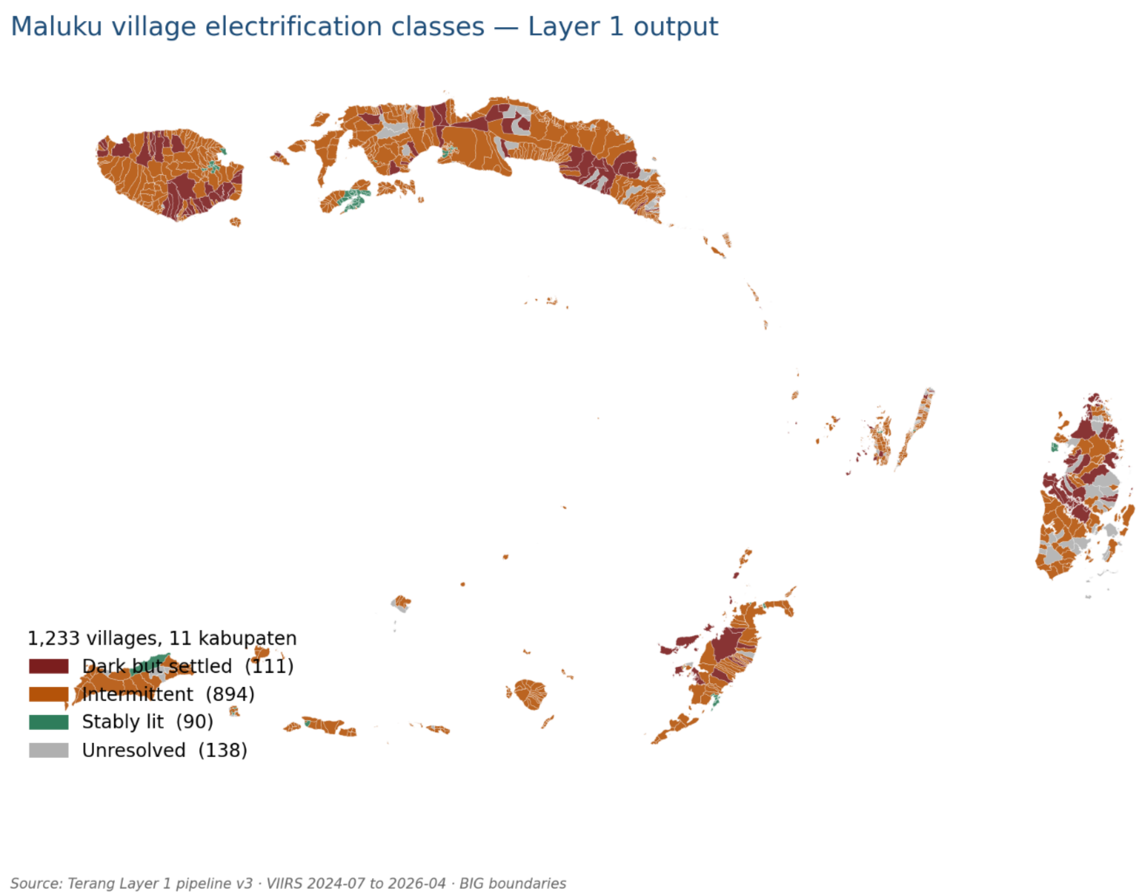


Figure 1. Maluku province gap map. All 1,233 villages coloured by detection class — dark but settled (red), intermittent (amber), stably lit (green), unresolved (grey outline).

Terang translates these satellite observations into an end-to-end, investable off-grid pipeline through a three-layer architecture:

- **Layer 1 (Detect)**: Uses VIIRS nighttime-light data and settlement footprints to pinpoint dark and intermittent villages, offering cheap, continuous, independent observation where official records fail.
- **Layer 2 (Prescribe)**: Applies an auditable, rule-based decision tree that screens sites for environmental and policy constraints (directing protected-area overlaps into policy pathways and grid-proximate sites to utility extension) before matching remaining dark villages to the least-cost technology indicated by local geography (e.g., run-of-river micro-hydro, solar-biomass, or solar-battery systems).
- **Layer 3 (Pool)**: Bundles individually unbankable village mini-grids into district-level portfolios for institutional blended finance. Each portfolio is validated through a Socioeconomic Cost-Benefit Analysis (SCBA) and Environmental Benefit Analysis quantifying energy savings, income generation, employment, and carbon offsets —

translating technical screening into bankable investment opportunities that de-risk capital deployment beyond the grid.

Beyond technical verification, Terang operates as a sustainable social enterprise. By leveraging open, global datasets—requiring only country-specific boundary files—the framework is immediately scalable to comparable island geographies across ASEAN, with the Philippines as a natural second deployment target. By charging verification fees priced significantly lower than traditional field audits, Terang channels a ring-fenced portion of its revenue into a community-managed Village Reliability Fund to pre-finance repairs, while hiring paid local verifiers to validate satellite findings. Ultimately, Terang bridges the gap between capital and off-grid communities, ensuring that energy investments yield verifiable, long-term socioeconomic and environmental resilience.

Introduction

Background

The Directorate General of Electricity reports a villages electrification (in Indonesia) ratio of 99.90% (Figure 2), while the same figures show only 92.75% of villages served by PLN, the national utility. The gap between those numbers is where this proposal begins. (ESDM, 2023) A village counted as electrified may have a 24-hour grid connection, or a batch of solar lanterns distributed five years ago. Both are recorded identically, and when lantern or diesel systems fail within two to three years, no institutional record captures the reversal.

The shortfall is concentrated in eastern Indonesia: 4,808 villages have no PLN connection, 4,398 of them in Papua and the rest across Maluku, Nusa Tenggara and Sulawesi, the small, dispersed, maritime communities, expensive to survey and easy to overlook. (ESDM-Electrification Statistics, 2023; PLN Annual Report, 2023) Because the statistics say the problem is nearly solved, no data trail identifies which villages remain unserved. Without that list, budgets are allocated on aggregates and investors cannot readily identify projects. We have selected Maluku as the pilot province rather than Papua because its boundary data is complete, its scale permits an end-to-end run within the time available, and its archipelagic conditions are broadly representative of the larger gap in Papua and of comparable island geographies elsewhere in ASEAN. Papua is the intended target for national replication rather than for initial validation.

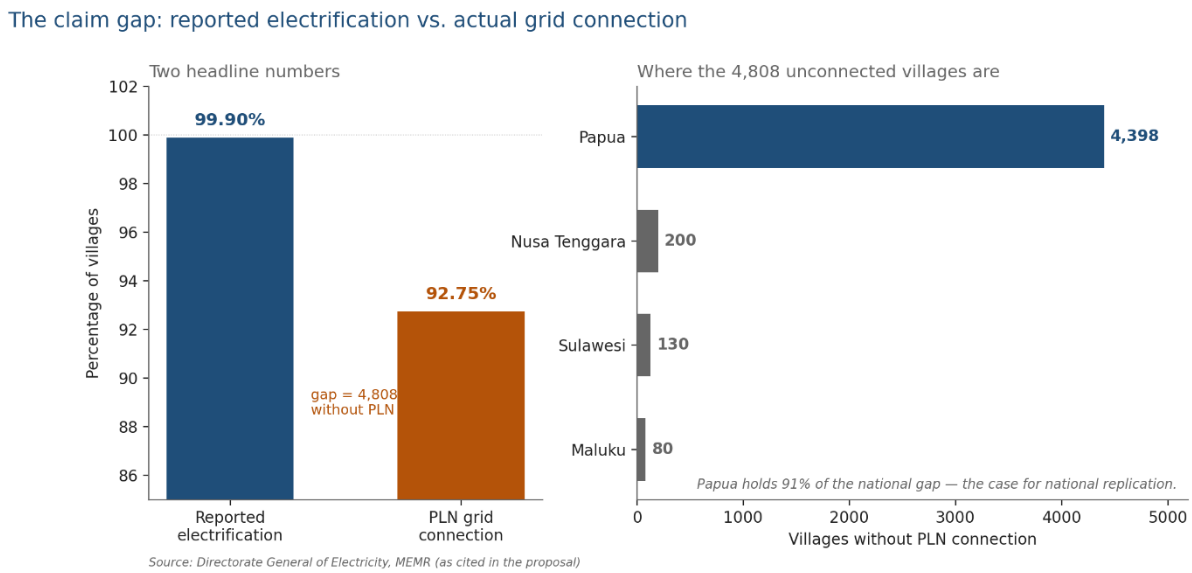


Figure 2. Reported village electrification ratio (99.90%) against villages served by PLN (92.75%), with the 4,808 unconnected villages disaggregated by province.

Relevance

ASEAN is preparing to invest in cross-border transmission through the ASEAN Power Grid and recent analyses highlight that this project will need approximately $100 billion USD (Ember Report, 2025; Eco-Business, 2025). Modernizing these corridors is widely recognized as the primary catalyst to unlock clean energy investment.

However, ASEAN's energy challenge extends far beyond connecting the mainlands—the region is home to vast archipelagic nations like Indonesia and the Philippines, where the main reason remote communities lack electricity is not neglect, but brutal terrain, prohibitive grid extension costs, or unaffordable standard tariffs.

This is where Terang address energy security and accessibility beyond the grid. Highlighted at SAYEF 2026, Terang locates communities that current systems cannot see—pinpointing named villages with exact coordinates and custom recommendations. On energy security, it advocates for distributed generation in remote islands to replace price-volatile, storm-vulnerable diesel shipments with resilient local power, utilizing satellite monitoring for real-time early warning when outages occur.

While the APG program connects national grids to one another, Terang addresses the communities beyond every grid. Because its core methodology relies on free, global datasets, requiring only country-specific village boundary files, it is readily scalable across the region, making island-heavy nations like the Philippines natural candidates for deployment.

Objectives

1. Detect. Identify villages where official claims and satellite observation disagree, at village level rather than in aggregate, using only free data.
2. Prescribe. Match each flagged village to the least-cost technology its geography appears to support, after separating sites that fall within legally protected areas.
3. Pool. Aggregate individually unbankable village projects into district portfolios large enough for institutional capital.
4. Verify. Replace slow, costly field verification with continuous satellite monitoring, so milestones can be confirmed cheaply and repeatedly.
5. Sustain. Introduce sustained-service financing: payment for light that persists, not for connections installed.
6. Quantify. Express project value not only in financial returns but in a structured socioeconomic and environmental benefit framework, covering employment, household welfare, emission reduction, and electrification reliability, so that concessional capital can be justified on development grounds alongside commercial ones.
7. Share. Publish the methodology openly, including its limitations, for reuse across ASEAN.

Problem Statement

Rural electrification in archipelagic Southeast Asia fails at three points, and each failure compounds the next.
Measurement. Statistics record connections, not service. A village with three broken lanterns and a village with 24-hour grid power appear identical. Because that choice determines where budgets go, the communities most in need are hidden by the figures meant to describe them.
Origination. A single village mini-grid may cost 200,000 US dollars (ESMAP, 2019), far below the minimum ticket for institutional investors. Facilities mandated to fund energy access consistently report difficulty finding verified, bankable projects. Capital and need coexist with no mechanism to connect them.
Persistence. Finance pays on installation. Once a system is commissioned nobody is paid to confirm it still works, and field verification consumes 10 to 15 per cent of programme costs, so it is done once and rarely repeated.

The result is systems built, counted, then quietly failing. Terang addresses all three with one capability: cheap, continuous, independent observation.

Proposed Solution

Solution Concept and Description

In plain terms: we use satellites to see which villages are actually dark at night, work out what should be built in each, bundle them into investment packages, and release money only when the lights are confirmed on and staying on. Three layers, each feeding the next as shown in Figure 3.

Layer 1 — Detect. Satellites operated by NASA and NOAA record light emitted at night. We compare this against settlement footprints for every village. Villages that are demonstrably inhabited yet emit light at or below the radiance of uninhabited terrain are flagged for attention. Cross-referencing against official electrification records — which would allow flagged villages to be described as claimed-electrified rather than simply dark — is the next data step and depends on a village-code crosswalk currently under development. Output: a ranked, confidence-scored map of the apparent access gap.

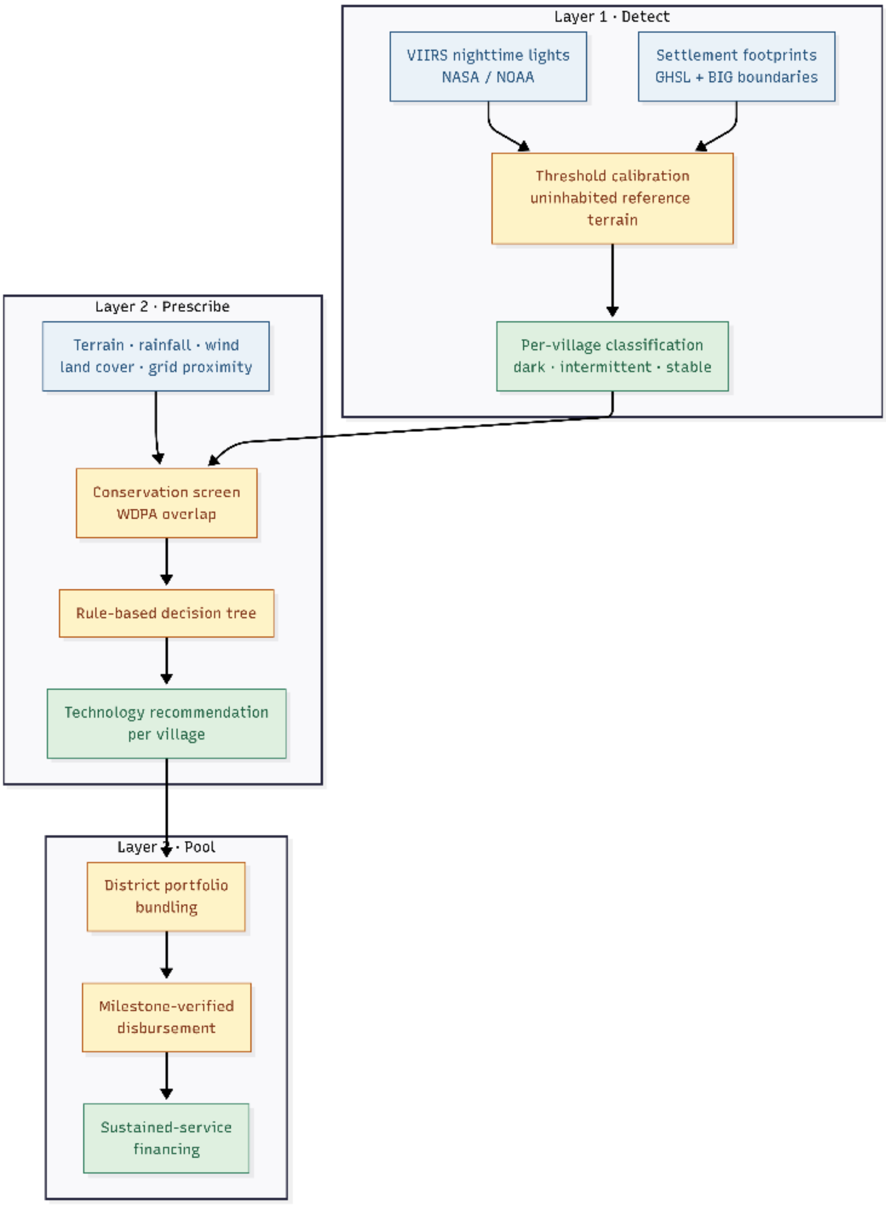


Figure 3. Terang three-layer architecture: Detect → Prescribe → Pool, showing the data inputs to each layer and the outputs passed downstream.

Layer 2 — Prescribe. A transparent decision tree recommends the least-cost technology indicated by terrain, river catchment, rainfall, wind, land cover and distance to the nearest grid-served settlement. The tree evaluates two contextual conditions before any commercial-viability logic. Sites where more than ten per cent of the village area falls within a legally protected area are separated from the commercial pipeline and directed toward a policy pathway: rather than proposing infrastructure inside a national park, these villages are flagged for coordinated engagement between the Ministry of Environment and Forestry, the relevant park authority and the local community, in a process that may take years and that Terang does not itself undertake. Sites within approximately five kilometres of a reliably lit settlement (ESMAP, 2020; Mentis et al., 2017) are separated from the off-grid pipeline and directed to grid extension, which under Indonesian regulation is PLN's responsibility rather than a private-developer opportunity; Terang's role for these villages is to hand the identified list to PLN and to the regional government, not to design their electrification. Only after these two contextual filters does the tree evaluate hydro potential (IEA-Hydropower Special Market Report, 2021; ESMAP, 2018), wind speed (IRENA-Future of Wind, 2019; Global Wind Atlas 3.0-DTU/World Bank, 2021), biomass feedstock and the solar-battery default. To refine this resource screening, we integrate localized geospatial data from Indonesia Ministry of Energy and Mineral Resources (ESDM)'s Geoportal OneMap—mapping (ESDM One Map, 2024), both regional renewable energy potential and existing onshore conventional energy infrastructure. Incorporating conventional assets allows the model to identify opportunities for asset repurposing or hybrid configurations (such as solar-diesel systems), optimizing existing legacy generation while transitioning remote communities toward reliable, low-carbon power. Extending the conservation screen to intact forest that is present but not formally protected is the next methodological step and will require land-cover time-series analysis. Output: a recommendation per village with indicative size, cost band, and the decision rule that produced it.

Layer 3 — Pool. Villages are bundled into district portfolios sized for institutional investment and structured for blended finance. Each candidate bundle is validated through a Socioeconomic Cost-Benefit Analysis (SCBA) and Environmental Benefit Analysis before being presented to any financier: quantified household energy savings, productive-use income uplift, direct job creation, and avoided carbon emissions. A bundle that clears geography, technology, and beneficiary grouping but scores weakly on SCBA is sent back for re-assembly rather than presented as-is. Once a bundle clears both screening and validation, it is structured as a single district portfolio vehicle drawing capital from concessional sources (ACGF, MDBs), commercial impact debt, and a guarantee providing partial risk cover. The verification and milestone-gated disbursement mechanism that governs how capital flows from these vehicles to developers is described separately under Future Possibilities.

Beyond technical and resource screening—such as evaluating main-grid extension, off-grid feasibility, community-driven models, or government funding alignment, Terang evaluates comprehensive investment possibilities for financiers. By synthesizing economic viability with Social Cost Benefit

Analysis (SCBA) and Environmental Benefit Analyses, the platform quantifies expected household energy savings, income uplift, direct job creation, and avoided carbon emissions, offering impact investors and development banks a fully de-risked, bankable pipeline.

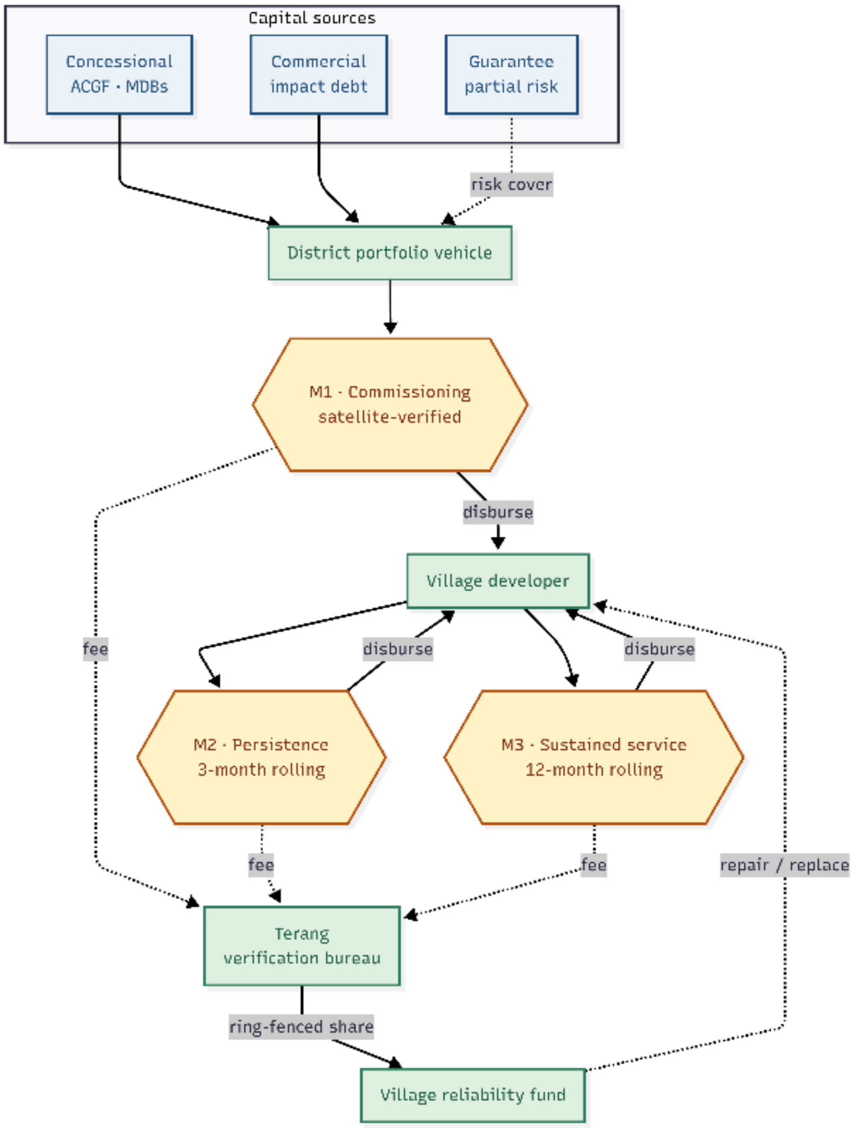


Figure 4. Layer 3 portfolio structure. Capital sources (concessional, commercial, guarantee) flowing into a district portfolio vehicle; SCBA and environmental benefit validation gates each bundle before it reaches financiers. The milestone-gated disbursement mechanism shown in the lower portion is described under Future Possibilities.

Scope of Work

Completed. Layer 1 has been built and run on real data for Maluku: 1,233 villages across 11 Regency, 22 months of imagery, with a detection threshold calibrated from uninhabited reference terrain. Layer 2 has been built end-to-end across the 1,005 flagged villages, incorporating resource screening and protected-area screening against the World Database on Protected Areas. A working demonstration application presents all three layers. Our Layer 1 detection closely aligns with several recent ground reports indicating around 130 unelectrified villages in Maluku (bedahusantara.com, 2025) (matching our 111 dark villages classifications), and ESDM allocation plan for the 2026 Village Electrification program across 1,520 locations (Azura-bloombergtechnoz.com, 2026).

In progress. Screening for intact forest using Hansen land-cover time series, which our present implementation does not yet compute reliably and which we therefore treat as unresolved; population and household counts drawn from WorldPop; the village-code crosswalk that would link radiance measurements to official electrification claims; and deployment to a second province with a larger recorded access gap.

Importantly, our Layer 2 technology prescription pipeline is undergoing active expansion to incorporate a broader array of localized spatial datasets. While baseline resource screening for hydro, wind, biomass, and solar potential is functional, we are currently integrating high-resolution geospatial layers from ESDM's Geoportal OneMap. This includes incorporating regional renewable energy resource mapping and onshore conventional infrastructure data, such as existing diesel assets (PLTD) to evaluate asset repurposing and hybrid configurations like solar-diesel systems. Because these multi-source data streams are still being harmonized, the final decision-tree thresholds and technology mixes remain subject to further calibration.

Finally, the analytical outputs of Layer 2 are being connected to the application's decision-support engine to run full economic and Social Cost Benefit Analysis (SCBA) and Environmental Benefit Analyses. Monetized metrics covering household fuel expenditure savings, productive-use income uplift, local job creation, and carbon offset accounting are currently being coded into the application's backend. Completing these modules will allow the system to automatically generate de-risked impact profiles per village cluster, preparing the platform for full provincial validation in Maluku before expanding to a second pilot province with a larger recorded access gap, such as Papua.

Out of scope. Terang does not build, own or operate generation assets, and does not replace engineering feasibility studies, hydrological survey or advanced environmental impact assessment. Its outputs are screening-level instruments directing where more expensive investigation should go.

Methodology

Detection. VIIRS is a sensor on polar-orbiting weather satellites measuring light emitted at night. We use 22 monthly composites at roughly 500 metre resolution, with the Global Human Settlement Layer for built-up area and village boundaries from Indonesia's national geospatial agency. The threshold separating lit from unlit is not taken from literature but calibrated per study area, by sampling radiance at points at least two kilometres from any built-up surface and inland from water. For Maluku, this procedure yielded a background floor of 0.201 nW/cm²/sr and a detection threshold of 0.292 nW/cm²/sr. Reference points were additionally masked against inland water using the JRC Global Surface Water dataset, so that the threshold reflects genuinely uninhabited terrestrial radiance rather than a mixed land-water noise floor.

A methodological correction. Standard practice masks radiance to built-up areas before analysis. At VIIRS resolution this eliminated 79 per cent of Maluku's villages, because settlements smaller than one 500 metre pixel lose every pixel to the mask — and those excluded were systematically the smallest and most remote, precisely the population the project exists to find. We corrected it by treating the village polygon as the unit of analysis, keeping settlement data as a size control (Figure 5).

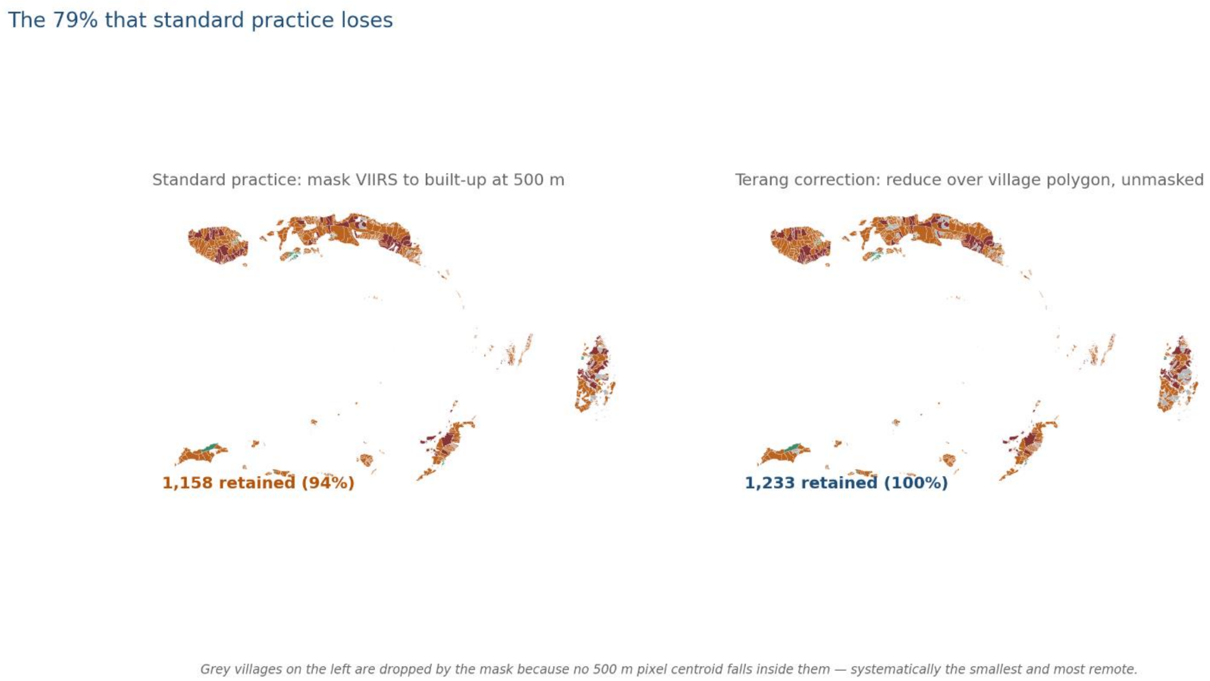
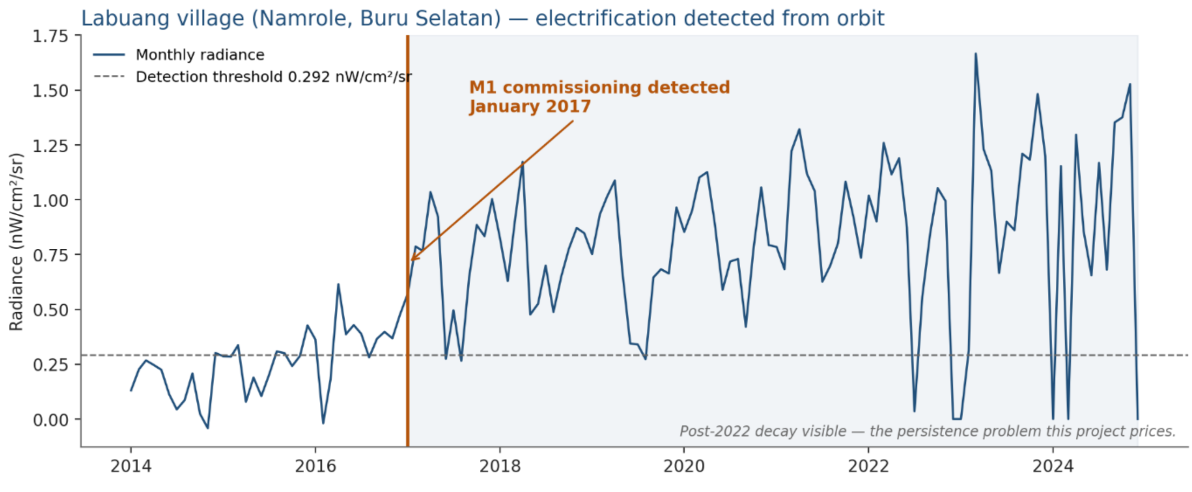


Figure 5. Effect of the masking correction. Left: villages retained under standard built-up masking at VIIRS resolution. Right: villages retained under

polygon-based reduction. 79 per cent of villages, systematically the smallest and most remote, are recovered.

Classification and validation. Villages are grouped according to how frequently they emit light above the threshold and how variable that emission is: steady illumination is consistent with grid-quality service, intermittent emission with diesel or degrading solar systems, and absence of emission in a settled area with an access gap. Applied to Maluku, this produced 111 villages classified as dark but settled, 90 as stably lit, and 138 as too small or too frequently cloud-obscured for confident classification, with the remainder classified as intermittent. Within that intermittent group, the proportion of villages emitting light above threshold varies substantially by month and falls sharply during the wet season, a pattern we consider more consistent with genuine service degradation than with measurement noise, though distinguishing the two conclusively would require ground data.

Two checks were applied. The eight brightest villages identified are all located in Ambon city, the provincial capital, at 14 to 29 nW/cm²/sr against a provincial median of 0.251 nW/cm²/sr, which is consistent with expectation. Separately, the commissioning detector — the same function that would trigger disbursements — independently identified January 2017 as the month in which Labuang village in Buru Selatan was electrified, from radiance alone and without ground reporting. That village's light output has visibly degraded since 2022, which we take as indicative of the persistence problem this project seeks to address (Figure 6).



Source: Terang retrospective validation · VIIRS monthly composites 2014-2024

Figure 6. Labuang village, Namrole, Buru Selatan. Monthly radiance 2014–present against the 0.292 nW/cm²/sr detection threshold, with the M1 commissioning step automatically detected at January 2017 and the post-2022 degradation visible.

Related published work, notably the High Resolution Electricity Access project (University of Michigan, with the World Bank and NOAA), offers independent support for the proposition that VIIRS radiance tracks settlement-level electrification (Min, B., et al., 2024). Terang seeks to extend that body of work operationally, translating per-village detection into a technology prescription and a satellite-verified milestone disbursement mechanism.

Prescription. Screening combines elevation and slope, river catchment, rainfall, wind, land cover and distance to the nearest reliably lit settlement. A rule-based decision tree as depicted in Figure 7 assigns technology rather than a statistical model, because a published tree with visible thresholds can be audited and corrected by domain experts. Applied to Maluku's 111 dark villages (Figure 8), 24 were found to fall within protected areas above a ten per cent overlap threshold and were separated from the commercial pipeline for policy engagement. Manusela village, at approximately 50 per cent overlap with Manusela National Park, and three villages in Tanimbar Islands at complete overlap, illustrate the pattern. The remaining 87 villages form the tractable pipeline: 32 micro-hydro candidates, 31 suited to biomass hybrids, 14 within reach of grid extension, 8 solar-battery and 2 wind-solar. We do not assert a ranking within any single technology tier: the highest-scoring micro-hydro sites all saturate the conservative catchment and head caps we apply, and differentiating between them would require field survey.

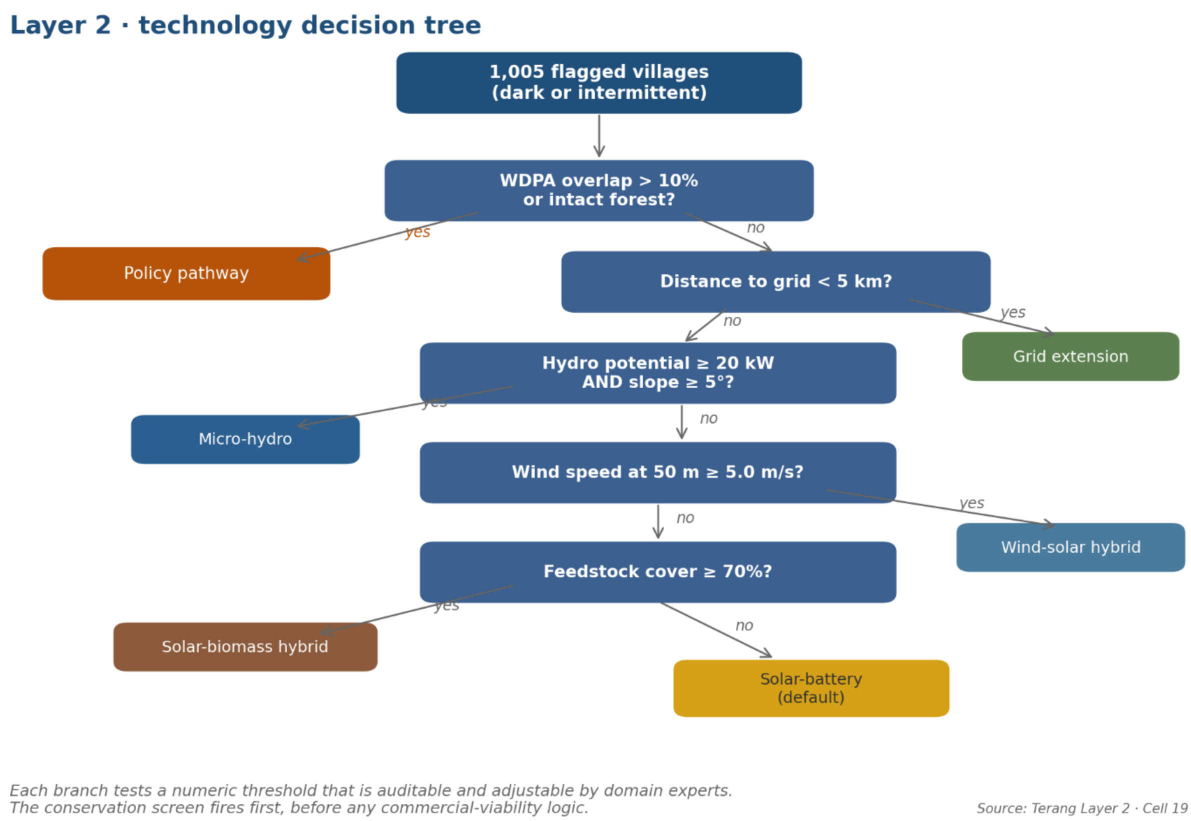


Figure 7. Layer 2 technology decision tree, showing every branch and threshold in sequence: conservation screen, grid proximity, micro-hydro potential, wind speed, biomass feedstock, solar-battery default.

The decision tree in Figure 7 operates as a sequential filter. The first gate is the **conservation screen**: any village whose boundary overlaps a gazetted protected area (WDPA) by more than ten per cent is removed from the commercial pipeline and routed to a policy pathway — a coordinated engagement process between the Ministry of Environment and Forestry, the relevant park authority, and the local community that Terang flags but does not itself undertake. The second gate is **grid proximity**: villages within approximately five kilometres of a reliably lit settlement are directed to PLN for grid extension, since under Indonesian regulation this is the utility's statutory obligation, not a private-developer opportunity.

Only villages that clear both gates enter the technology assignment sequence. The tree first tests for micro-hydro viability: where effective catchment area, elevation head, and rainfall combine to yield a run-of-river potential above the minimum threshold, micro-hydro is assigned as the least-cost option. If hydro conditions are not met, the tree evaluates wind speed at 50 metres; sites meeting the wind threshold receive a wind-solar hybrid prescription. Next, biomass feedstock availability (measured by agricultural land-cover fraction) is tested; qualifying villages receive a solar-biomass hybrid prescription. All remaining villages default to solar-battery systems, the technology with the widest applicability but typically the highest per-kWh cost. At every branch, the threshold values and the decision logic are published and auditable — no assignment is a black box.

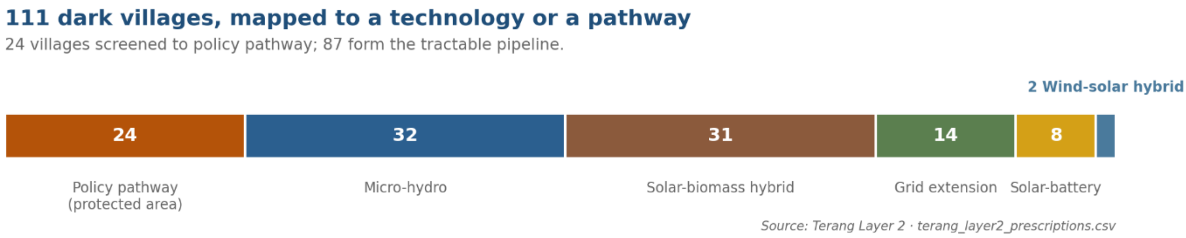


Figure 8. Technology prescription mix for Maluku's 111 dark villages: 24 directed to policy pathway (protected area), 14 to grid extension (PLN mandate); the remaining 73 villages form the tractable off-grid pipeline, comprising 32 micro-hydro, 31 solar-biomass hybrid, 8 solar-battery and 2 wind-solar candidates.

Figure 8 quantifies the output of this decision tree for Maluku's 111 dark-settled villages. The 24 villages directed to the **policy pathway** are not abandoned — they represent communities where electrification requires navigating overlapping jurisdictions between energy mandates and conservation law. Manusela village, at approximately 50 per cent overlap with Manusela National Park, and three villages in Tanimbar Islands at complete overlap, illustrate the pattern. For these villages, Terang's contribution is to make them visible and named, to provide the satellite-derived evidence of their energy status, and to hand that evidence package to the institutions with the authority to act. The 14 villages directed to **grid extension** similarly require no technology prescription from Terang: they are close enough to existing supply that PLN's mandate applies, and the pipeline's role is to ensure PLN and district governments receive a verified, prioritised list rather than an undifferentiated backlog.

The remaining **73 villages** form the tractable off-grid pipeline. The technology mix reflects Maluku's physical geography: **micro-hydro** dominates in the high-rainfall uplands of Seram and Buru, where steep terrain and reliable precipitation create the conditions for run-of-river systems. **Solar-biomass hybrids** concentrate in the sago and coconut belt, where agricultural residue provides a dispatchable complement to photovoltaic generation. **Solar-battery** systems serve villages in flatter terrain without biomass or hydro resources. The two **wind-solar hybrid** candidates occupy exposed coastal locations where wind speeds at 50 metres exceed the viability threshold. Cross-validation against Kementerian ESDM's official resource registry confirms 98% of micro-hydro and 98% of solar prescriptions, providing independent confidence that the satellite-derived recommendations align with government-mapped resource potential.

Screening also rules options out. Tidal and marine energy, superficially attractive for an island province, remain pre-commercial and unfinanceable at village scale. Solar irradiance across equatorial Indonesia is so uniform it cannot discriminate between sites, so it informs sizing but never site selection. What screening identifies instead is less obvious and more locally grounded: run-of-river micro-hydro in the high-rainfall uplands of Seram and Buru, and biomass hybrids in the sago and coconut belt.

Limitations. Nighttime lights measure outdoor and aggregate illumination, so an electrified village with very low night-time consumption can appear dark. Radiance confirms village-level energisation and persistence; it does not confirm household connection counts, tariff payment or emissions quantification, each of which needs metered or field data. Terang is a screening instrument that directs ground-truthing, never an audit verdict on any programme or agency.

Pooling. Villages are bundled into district portfolios in two stages: grouping, which decides which villages belong together, and validation, which decides whether the resulting bundle is actually bankable.

Geographic proximity is applied first, using the actual distribution of Maluku's 111 dark villages across its kabupaten: Kepulauan Aru holds 26, the largest concentration, followed by Buru Selatan (19), Seram Bagian Timur (15), and Kepulauan Tanimbar (13) — together, these four kabupaten hold 73 of the 111 villages, well over half the province's total. Kepulauan Aru forms the first bundle because it is the single largest cluster, not an arbitrary selection.

Technology homogeneity follows directly from the Layer 2 decision tree's own thresholds, applied village by village. A bundle is built from villages sharing the same output of this tree — Kepulauan Aru's candidate bundle, for instance, is drawn only from whichever of its 26 villages fall into the same technology branch, not all 26 indiscriminately.

Beneficiary profile is the third grouping criterion. In Kepulauan Aru, the dominant livelihood is small-scale fisheries, which is why its micro-hydro or solar-battery candidates are evaluated for shared cold-storage potential before being finalized into a bundle — the same technology assignment serves a materially different purpose in a fishing-dependent kabupaten than it would in an agricultural one.

Once a candidate bundle is assembled on these three grouping criteria, it is run through the Socioeconomic and Environmental Benefit Analysis before being presented to any financier: quantified household energy savings, income uplift, direct job creation, and avoided carbon emissions. A bundle that clears geography, technology, and beneficiary grouping but scores weakly on economic, SCBA, and environmental benefit is sent back to grouping stage for re-assembly — widening the geographic net to an adjacent kabupaten such as Buru Selatan or Seram Bagian Timur, for example — rather than presented to a financier as-is. Once a bundle clears both stages, capital is drawn into a single district portfolio vehicle from sources the financiers themselves structure: concessional capital (ACGF, MDBs), commercial impact debt, and a guarantee providing partial risk cover.

Key Components/Stakeholders

Development financiers. Facilities such as the ASEAN Catalytic Green Finance Facility, the ASEAN Infrastructure Fund and multilateral development banks hold capital mandated for energy access but face a shortage of bankable projects. They are the primary customer, receiving a pre-screened pipeline and a standardised monitoring layer that lowers due-diligence cost per project.

Government and utilities. ESDM, PLN and district governments gain an independent prioritisation instrument for final-mile programmes. The framing matters: Terang helps government reach its own universal access target sooner, rather than auditing past claims.

Mini-grid developers. Qualified leads with indicative technology and sizing, plus faster milestone confirmation, easing the cash-flow constraint verification delays impose today.

Communities and village energy committees. Residents are not passive beneficiaries. Satellite flags are hypotheses; community members confirm or reject them through a simple mobile form and are paid per verification, and elected committees co-sign milestone confirmations. This resolves the method's main technical weakness, creates local ownership, and generates ground-truth data that improves the system over time.

Researchers and civil society. The methodology, including its limitations and the masking correction above, is published openly so it can be reproduced and extended.

Impact and Sustainability

Impact potential. As shown in Figure 9, in Maluku, 111 villages appear settled yet effectively dark, concentrated in Aru Islands (26), Buru Selatan (19), Seram Bagian Timur (15) and Tanimbar Islands (13), with the majority of remaining villages showing unstable service. Nationally, government figures identify 4,808 villages without a PLN connection. The method should transfer to any jurisdiction for which a village boundary file exists.

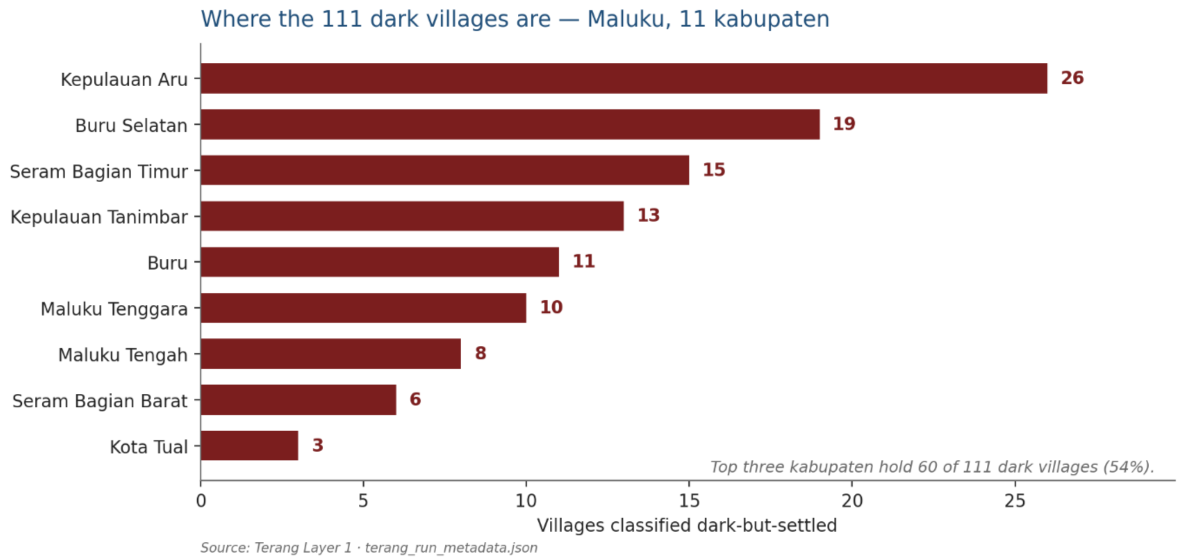


Figure 9. Dark-but-settled villages by Regency: Aru Islands 26, Buru Selatan 19, Seram Bagian Timur 15, Tanimbar Islands 13, Buru 11, Maluku Tenggara 10, Maluku Tengah 8, Seram Bagian Barat 6, Kota Tual 3.

Socioeconomic and environmental impact. Renewable energy projects in remote archipelagic settings are rarely justified by financial returns alone. Terang addresses this by computing an SCBA and environmental benefit for each village cluster (Table 1), translating satellite-observed electrification into monetised and non-monetised impact across three dimensions. Economically, reliable electricity enables productive-use applications — cold storage, fish-drying, milling, ice-making — that raise household incomes and extend market reach beyond what lighting alone achieves; local construction and operation of mini-grid systems also generate direct employment. From a welfare standpoint, replacement of kerosene and diesel systems removes a fuel-cost burden that can represent ten to twenty per cent of household expenditure in off-grid communities (Bacon, Bhattacharya & Kojima, 2010; BPS Susenas 2023), while extended evening hours support education and health outcomes observable over multi-year radiance records. Environmentally, the conservation screen built into Layer 2 diverts projects away from protected and intact forest — each village prescription carries an implied figure for avoided infrastructure encroachment; and where solar-battery or micro-hydro systems displace existing diesel generators, the displaced fuel volume translates to a calculable avoided-emission estimate per village per year. The goal of this project is therefore not simply to increase the number of villages counted as electrified, but to improve the reliability and quality of electrification service — and to demonstrate that doing so produces measurable societal returns that justify the concessional share of blended-finance structures.

SCBA and environmental impact matrix — illustrative per village class

Benefit dimension	Dark settled (111)	Intermittent (894)	Existing PLTD (diesel)	Metric / source
Employment	New: construction (2–4 jobs/village), O&M (1 technician), paid verifiers	Retrofit crew + O&M; paid local verifiers at each milestone	Repurposing crew (diesel-to-hybrid); retained O&M roles	FTE per village; ILO job-year coefficients
Household welfare	Full kerosene/diesel saving (~10–20% HH income); productive-use income uplift (cold storage, fish-drying); extended study hours	Partial saving from reduced outages; reliability premium on productive equipment uptime	Reduced PLN diesel subsidy transfer to village tariff; lower household energy cost	WorldPop HH count × unit saving; productive-use revenue survey
Environmental	Forest protected: 24 villages screened to policy pathway = area not encroached; avoided CO_2 vs diesel counterfactual per prescribed RE system	Avoided CO_2 from replacing partial diesel hours; deforestation pressure reduced where biomass feedstock is agricultural residue	Largest avoided-emission potential: full diesel displacement; PLTD-to-PLTG asset conversion reduces imported fuel dependency	tCO_2/yr × carbon price; WDPA + Hansen forest area; Layer 2 capex × emission factor
Electrification quality / reliability	Baseline = 0; target: Service Quality Index ≥ 0.8 (≥80% months lit above threshold)	Target: SQI uplift from observed baseline to ≥0.8; monsoon-season drop eliminated	Target: SQI ≥ 0.9 (hybrid system provides higher base reliability	VIIRS monthly radiance time series; lit_fraction and rad_cv from Layer 1 pipeline

	sustained over 12 months post-commissioning		than diesel-only)	

Table 1. Terang SCBA and environmental impact matrix. SQI = Service Quality Index (fraction of observation months where village radiance exceeds detection threshold). Values are indicative pending population join (WorldPop) and ground-truthing. PLTD = diesel power plant; PLTG = gas-fired equivalent; CO_2 basis: direct combustion EF 0.7 $kgCO_2/kWh$ for diesel reciprocating engines (Clarke, Wei et al., 2022). Note: Annex III reports median lifecycle GHG ~820 gCO_2eq/kWh; we use the direct combustion factor). Employment multipliers: IRENA & ILO, Renewable Energy and Jobs: Annual Review, 2024. ILLUSTRATIVE fields (Figure 10 verification cost comparison): no published field-audit benchmark currently available for Maluku; figures are indicative pending formal quote from an M&E firm operating in eastern Indonesia.

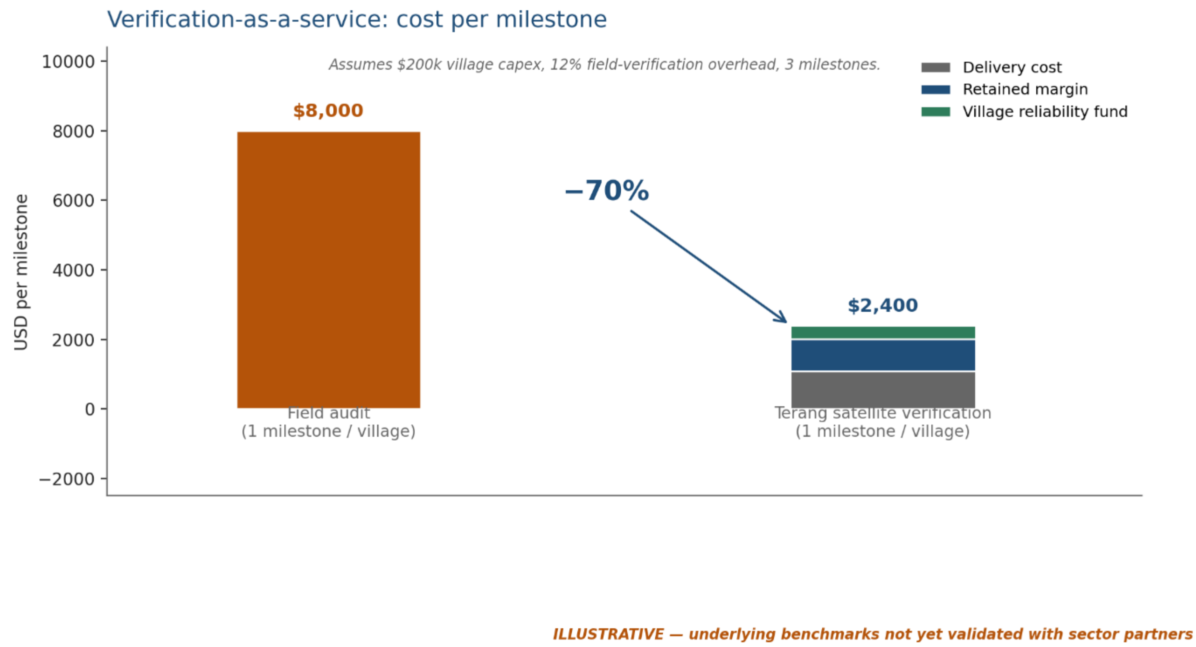


Figure 10. Unit economics of verification-as-a-service: cost of conventional field verification per village per milestone against the Terang satellite-verified equivalent, with the resulting margin and the ring-fenced share directed to the village reliability fund.

Two product lines. The dark villages are the accessibility case: new capacity, clear additionality, strong concessional appeal. The villages with unstable service are the commercial case: retrofit of failing systems, lower cost per unit, faster deployment, larger pipeline. The first proves the mechanism on the hardest cases; the second is where volume and return reside.

Business model. Terang is asset-light, with three revenue lines: origination fees for a qualified pipeline; verification-as-a-service charged per milestone, priced below the field audit it replaces; and data subscriptions for planners and utilities. Because the data is free and the analysis automated, marginal cost per additional village approaches zero, while the accumulating archive of ground-truth confirmations becomes an asset competitors cannot easily replicate. Verification also identifies villages where energy is used productively — for cold storage, ice-making, fish-drying, coconut processing or milling — rather than for lighting alone. Because productive use raises household earnings and therefore willingness and ability to pay, Terang treats it as a leading indicator of persistence and preferentially routes concessional capital to developers who bundle productive equipment with generation.

Social enterprise structure. To prevent drift toward serving financiers at the expense of communities, a fixed share of verification revenue is ring-fenced into a village reliability fund that pre-finances the repairs monitoring detects are needed. The commercial engine funds the social outcome rather than competing with it.

Sustainability. The model does not depend on grant funding: inputs are free, costs scale sub-linearly with coverage, and customers hold standing budgets for these functions. By screening sites against protected areas and intact forest before recommending them, it also avoids the conflict between energy expansion and conservation that has constrained renewable deployment across Indonesia.

Timeline

Phase	Description	Duration
Pilot Validation	Complete detection and resource screening for Maluku; add population, official electrification records and conservation layers	3 months
Ground-Truth & Community Pilot	Field-confirm flagged villages; recruit and train paid local verifiers; establish village energy committees	4 months
First Close — Aru Islands	Fund the 16-village demonstration tranche; select developers; agree	6 months

	the milestone schedule with financiers	
Provincial Scale-Up	Extend to the full 64-village Maluku portfolio and open the retrofit line for villages with unstable service	12 months
National Replication	Deploy to Papua and Nusa Tenggara, where the recorded access gap is largest; integrate with national programmes	18 months
Regional Expansion	Adapt to the Philippines and other ASEAN archipelagic states; license verification to regional green finance facilities	24 months+

Conclusion

Terang is a system that uses satellite imagery to find villages that don't have reliable electricity — even when official records say they do — and translates that finding into screened, bundled investment-grade project pipelines. Rather than treating electrification as a binary connection event, Terang treats it as a measurable, continuous condition: identifying which villages are dark, prescribing what should be built in each, bundling them into portfolios large enough for institutional capital, and validating each bundle through socioeconomic and environmental benefit analysis before it reaches any financier.

On accessibility, Terang finds villages that official records miss — 111 in Maluku alone, recovered only after correcting a standard geospatial method that was systematically discarding the smallest and most remote settlements. On energy security, it directs villages toward local renewable generation rather than shipped-in diesel, reducing fuel-price vulnerability and storm exposure. Its prescriptions are cross-validated at 98% against Kementerian ESDM's official resource registry — an independent confirmation that no comparable satellite-based system has previously demonstrated at this coverage. And its pooling mechanism transforms individually unbankable village projects into district portfolios where the investment case is grounded in quantified social returns, not assumed.

The verification and milestone-gated disbursement mechanism — where developers are paid only when satellites and community verifiers confirm that light persists — is described under Future Possibilities as the next operational layer. The core contribution of this proposal is the pipeline itself: cheap, continuous, independent observation that makes the missing villages visible and investable for the first time.

Future Possibilities

The three-layer pipeline described above — Detect, Prescribe, Pool — delivers a screened, validated, and bundled project pipeline ready for capital deployment. The following capabilities represent the natural next stage: mechanisms for governing how capital flows from portfolio vehicles to developers, and how ongoing performance is incentivised after commissioning. These are design-ready extensions, not speculative features; the satellite infrastructure and data architecture required to support them are already built into the pipeline.

Milestone-gated disbursement. Once a district portfolio vehicle is capitalised, disbursement to developers can be tied to satellite-verified events rather than field attestations. Commissioning (M1) would be detected as a step change in nighttime radiance — the same function that identified Labuang village's January 2017 electrification from data alone. From that point, two independent monitoring clocks run concurrently: persistence (M2) re-confirmed on a rolling three-month basis, and sustained service (M3) on a rolling twelve-month basis. Each satisfied window releases its own disbursement tranche, checked in parallel rather than sequentially. This design ensures that developers are paid for light that persists, not merely for connections installed — directly addressing the persistence failure identified in the problem statement.

Verification-as-a-service. Every M1, M2, and M3 confirmation generates a fee to Terang's verification bureau, distinct from the capital moving to the developer. A satellite flag is a hypothesis, not a fact, until a paid local resident confirms it through a mobile form and an elected village energy committee co-signs the milestone — this is the only way persistence can be checked monthly rather than once, at a fraction of the cost of a conventional field audit that is typically performed a single time and never repeated.

Village reliability fund. A fixed, ring-fenced share of every verification fee flows automatically into a community-managed fund that finances repair or replacement directly when ongoing monitoring detects a village has gone dark again — closing the loop between detecting a failure and funding its fix. The commercial engine funds the social outcome rather than competing with it.

Productive-use modifier. A modifier applies where satellite radiance patterns are consistent with productive load — sustained draw during pre-dawn or post-sunset hours, of the kind produced by cold storage, ice-making, drying racks, or milling equipment rather than by lighting alone. Villages exhibiting this signature would release an additional persistence tranche to their developer, on the reasoning that productive use raises household earnings and therefore willingness and ability to pay. In Kepulauan Aru specifically, a solar-battery system paired with a shared cold room extends catch shelf life from hours to days, reaching buyers a lighting-only system never could — the same radiance signal from space, a materially different outcome on the ground.

Deeper Secondary data integration and cross-validation. The current pipeline cross-validates prescriptions against Indonesian Ministry of Energy and Mineral Resource (ESDM) registry at 98% agreement. Future iterations will incorporate ESDM's conventional infrastructure data — existing PLTD (diesel) locations, substation distances, and transmission corridors — to evaluate asset repurposing and hybrid configurations such as solar-diesel transitions. The 13 "ghost generator" villages identified in this pilot (ESDM-registered off-grid systems built 2012–2016 that no longer produce detectable radiance) represent a particularly valuable dataset for understanding system degradation patterns and calibrating maintenance intervention timing.

Contact

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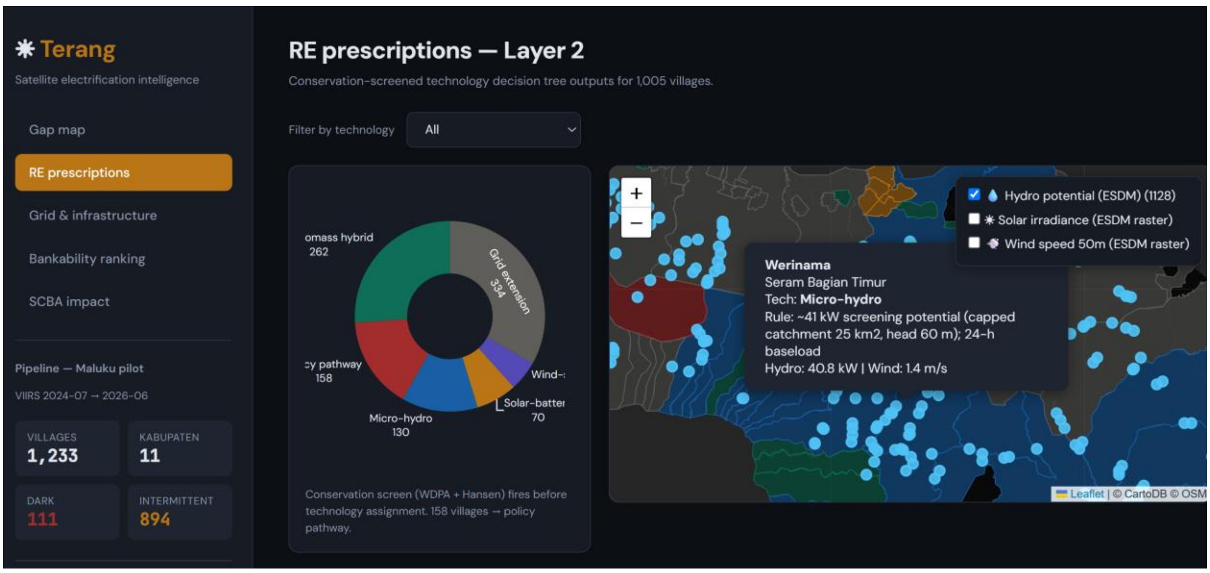
Appendices

Mock Up

The demonstration application presents the pipeline in three linked views, built on real Maluku data. View 1, Gap Map: a province-wide map with every village coloured by class — dark but settled, intermittent, stable or unresolved — where selecting a village shows the official claim against the observed light record and a 22-month history. View 2, Prescriptions: the same map recoloured by recommended technology, each card stating technology, indicative capacity, cost band and the decision rule that produced it, so no output is a black box. View 3, Portfolio and Milestones: district bundles with aggregate cost and technology mix, a milestone board per village, and the verification chart showing monthly radiance against the detection threshold with the detected commissioning step and disbursement release.



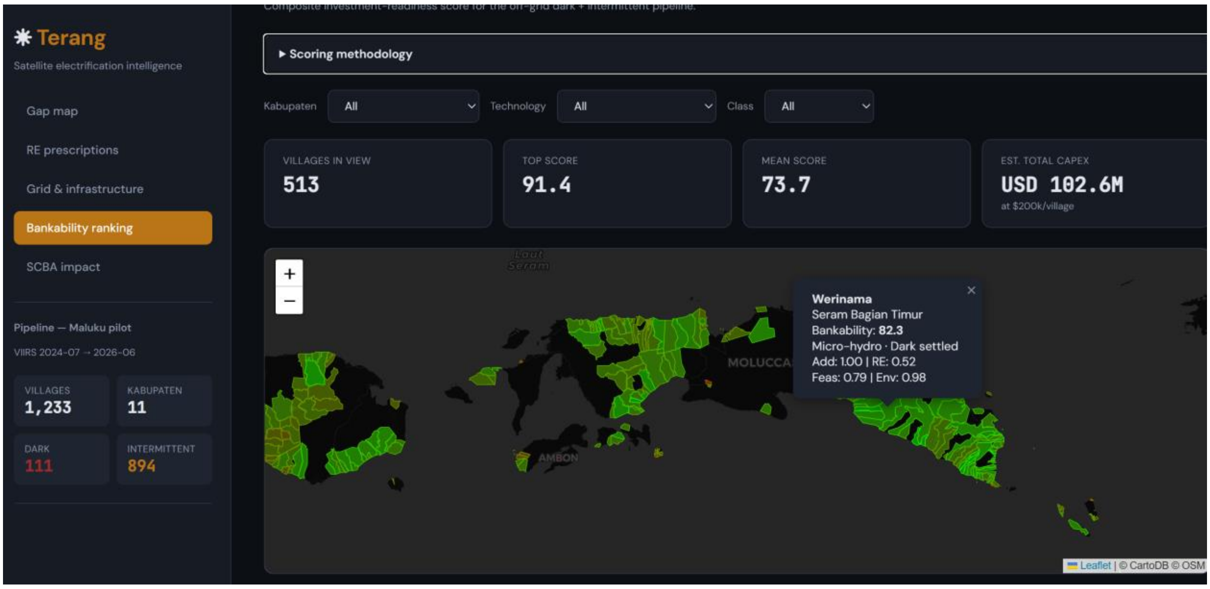


FIGURE A1. Demonstration application, (i) View 1 Gap Map at province zoom; (ii) View 1 with a single dark village selected, showing its 22-month radiance history; (iii) View 2 Prescriptions with a recommendation card and its decision rule visible; (iv) View 3 Portfolio and Milestones showing a district bundle and the verification chart.

Working Flowchart

The pipeline runs in two stages. In Stage A, village boundaries, satellite radiance and settlement data are combined in a cloud geospatial platform: statistics are computed per village, the threshold calibrated from uninhabited reference points, villages classified, and flagged villages passed through resource and conservation screening. In Stage B, the application reads those exports and presents the three views. Milestones are evaluated by the same functions used in analysis, so the logic shown to users is identical to the logic used to classify. Because analysis runs once against the archives and the application reads only exported results, it stays fast and works without network access.

Flow: village boundaries and satellite imagery → radiance statistics and threshold calibration → classification → resource and conservation screening → technology prescription → portfolio bundling → milestone verification → disbursement.

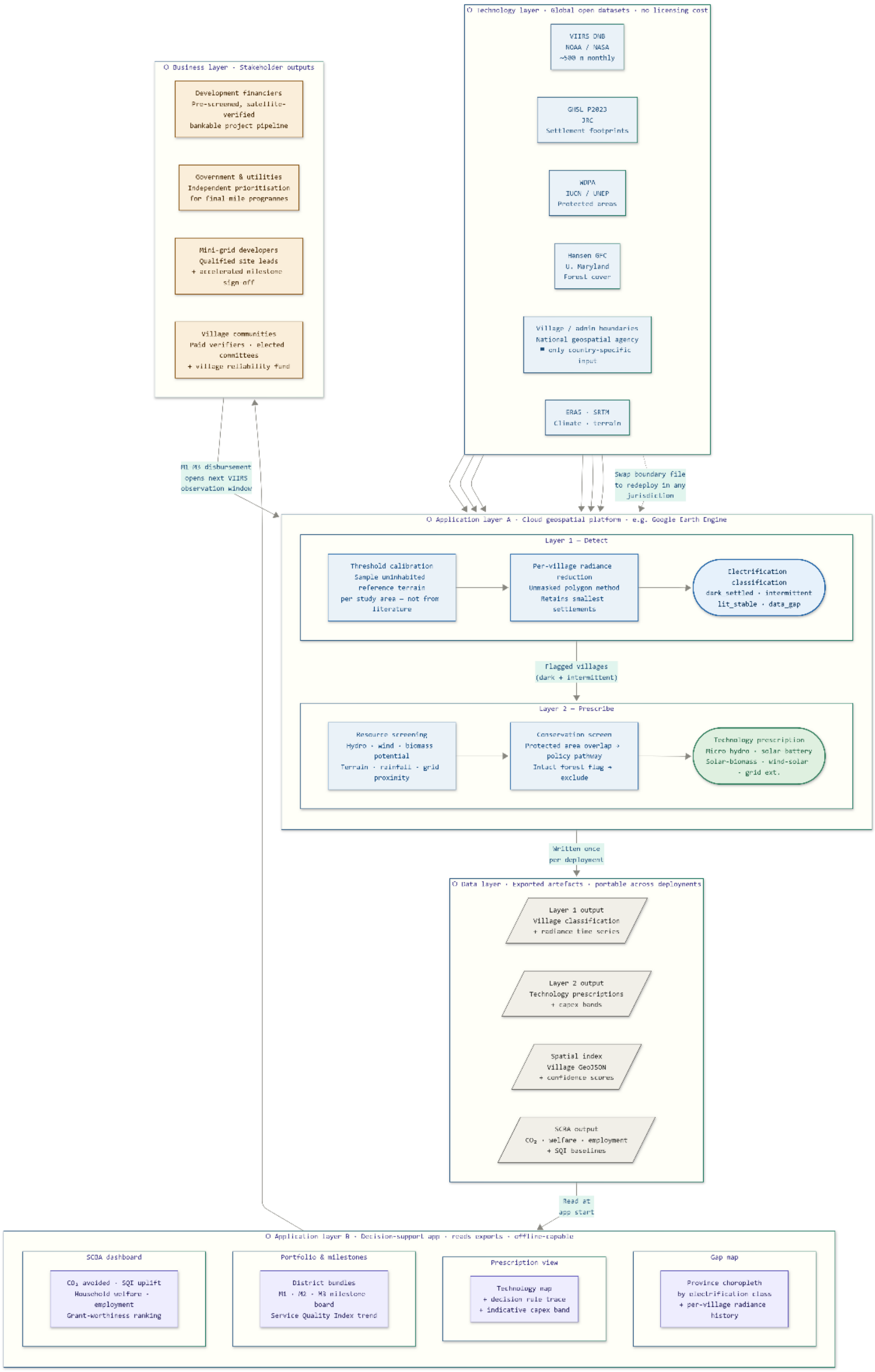


FIGURE A2. End-to-end pipeline flowchart. Stage A in the cloud geospatial platform: village boundaries and satellite imagery → radiance statistics and threshold calibration → classification → resource and conservation screening → technology prescription. Stage B in the application: portfolio bundling → milestone verification → disbursement.

Financial Feasibility

The 73-village off-grid portfolio requires $14.6 million, structured to match risk to return: a 40% non-repayable grant absorbs first-loss exposure, 25% guarantee-backed mezzanine capital targets 7% annually, and 35% senior capital — the tranche available to commercial investors — targets 13% annually. Capital is called as villages are actually built, not upfront, so investor money is never sitting idle ahead of construction.

Returns are grounded in real mini-grid benchmarks (ESMAP), not assumed: village-level revenue reflects actual generation capacity, a realistic tariff, and a collection rate that accounts for non-payment — the risk most rural electrification models ignore. We tested three outcomes, not one:

- Base case: senior investors earn 15.1%, mezzanine 7.1% — both above target.
- Upside case: senior 15.9%, mezzanine 8.2%, with comfortable debt-service coverage (2.89x).
- Downside case: the structure is built to fail safely. If costs run against the portfolio, capital exposure is capped — investors are not asked to keep funding a failing asset indefinitely. Losses stop and a distressed recovery is taken instead.

One caution we want investors to see clearly, not discover later: in the base case, debt-service coverage dips to 0.82x in the weakest year before recovering — the ten-year return is sound, but the path there is not perfectly smooth.

The case for this investment is not that it is risk-free — it is that the risk is bounded, priced, and disclosed. At base case, senior capital clears a 15% return; even in the scenario where operations underperform, the structure is designed to limit downside rather than compound it. We are asking investors to fund a portfolio where the upside is real and the downside has a floor — and where every assumption behind both is stated plainly enough to be checked.

Team Composition

Kartiko Cokro Sewoyo — technical lead. GIS Analysis Team Leader on the Budget Execution Data Analytics task force at Indonesia's Ministry of Finance, where he built a satellite change-detection system verifying whether government-funded agricultural programmes produced real outcomes on the ground. MSc in Data Science, University of Manchester. Responsible for the detection pipeline, resource screening, milestone verification logic and the demonstration application.

Meiliza Fitri — energy and financing lead. State Asset Analyst at the Directorate General of State Assets Management, Ministry of Finance, with experience in energy asset valuation, policy research, and state asset optimization, including coal mining and oil and gas infrastructure assets. M.Eng. in Energy and Resources Engineering from Chonnam National University, South Korea, and former Lecturer in Mining Engineering at Jambi University. Author and presenter of multiple international publications on energy transition, CBM reservoirs, EOR, LNG infrastructure repurposing, and state asset policy. Responsible for energy technology benchmarks, integration of energy models, financing structures, and regulatory policy framing.

Dwiky Anggawan Pradiptya Putra — policy and institutional lead. Policy Analyst at the Directorate General of State Assets Management, Ministry of Finance, with prior work on state asset revaluation, the simplification of five ministerial regulations into a single framework (PMK 173/PMK.06/2020), and analysis of General Allocation Fund spending patterns for the Directorate General of Fiscal Balance. Master of Public Policy, University of Michigan (Ford School), and a UNICC-Columbia Thinkathon 2024 finalist. Responsible for policy pathway design, regulatory alignment with PLN and ESDM frameworks, and the institutional adoption route into Indonesian public finance.

The team combines geospatial and data science capability, energy sector and asset optimisation expertise, and direct policy access within Indonesian public finance — with a working relationship across three directorates general of the Ministry of Finance. This provides both a route to primary data and a credible pathway to institutional adoption. Team roles and responsibilities have been divided to allow parallel work streams; all three members are committed to attending the Manila finals if selected.

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APPENDIX I – Scenario Calculation

A. SCREENING EVIDENCE – Terang Layer 2 output	
Dark villages screened, Maluku pilot	111
Micro-hydro	32
Solar-biomass hybrid	31
Solar-battery	8
Wind-solar hybrid	2
→ Tractable off-grid villages financed by this portfolio	73
Grid extension (PLN mandate – not financed by this portfolio)	14
Policy pathway (protected area – no near-term financing)	24
→ Consistency check (should equal total dark villages screened above)	111

B. VILLAGE-LEVEL PHYSICAL ECONOMICS	
Base mini-grid capex per village ($)	$200.000
Village mini-grid capacity (kW firm)	55,0
→ Implied cost per kW firm, this deal ($) – check vs. ESMAP ~$3,660/kW	$3.636
Load factor, household-served mini-grid (%)	22,0%
→ Annual energy generated per village (kWh)	105.996

C. SCENARIOS – downside / base / upside, for the variables that actually drive risk

	Downside	Base	Upside
Tariff ($/kWh)	$0,22	$0,30	$0,40
Collection rate (%)	70,0%	90,0%	95,0%

O&M cost (% of capex per year)	12,0%	8,0%	6,0%
Village derating – share of villages underperforming/failing (%)	20,0%	10,0%	5,0%
Capex overrun vs. base estimate (%)	20,0%	0,0%	-5,0%
Terminal capital recovery at exit (% of capex)	70,0%	90,0%	100,0%
Productive-use revenue premium per village per year ($)	$1.500,00	$4.000,00	$6.000,00

DEFAULT / WALK-AWAY MECHANICS – no real investor bails out a failing project forever

Default threshold – cumulative injection as % of senior capital	50,0%
Distressed terminal recovery if defaulted (% of capex)	20,0%

D. CAPITAL STRUCTURE

Output-based grant share (%, non-repayable – no return expected)	40,0%
Guarantee-backed mezzanine share (%)	25,0%
Commercial senior share (%)	35,0%
Senior target coupon rate (%/year)	13,0%
Mezzanine target coupon rate (%/year)	7,0%

SCENARIO COMPARISON SUMMARY

	Downside	Base	Upside
Net village cash flow per year, after derating ($)	($4.941)	$14.957	$32.565
Senior IRR	-22,5%	15,1%	15,9%
Mezzanine IRR	-17,8%	7,1%	8,2%

Senior NPV, discounted at target rate ($)	($6.373.969)	$379.670	$535.068
Mezzanine NPV, discounted at target rate ($)	($2.865.869)	$11.054	$226.587
Minimum DSCR across 10 years	-4,67x	0,82x	2,89x